

Humanoid World Model

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Introduction

Motivation - The Need for World Models

Do We Have Enough Data For Humanoid Robot?



Do We Have Enough Data For Humanoid Robot?

Usually we use real world data,
language, vision, audio ...

But is that enough for robots?

Do We Have Enough Data For Humanoid Robot?

Datasets

 **1M Episodes** from **311 Scenes** 
 **34 Research Labs** across **21 Institutions** 

22 Embodiments



527 Skills



60 Datasets

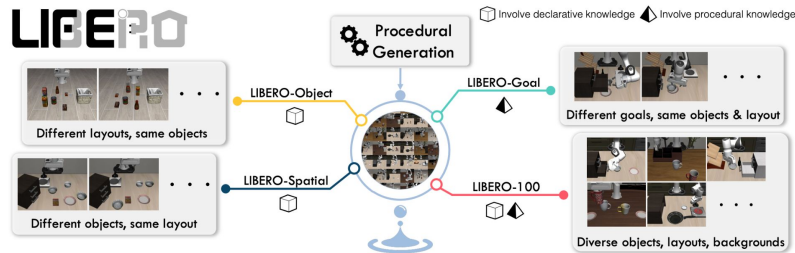
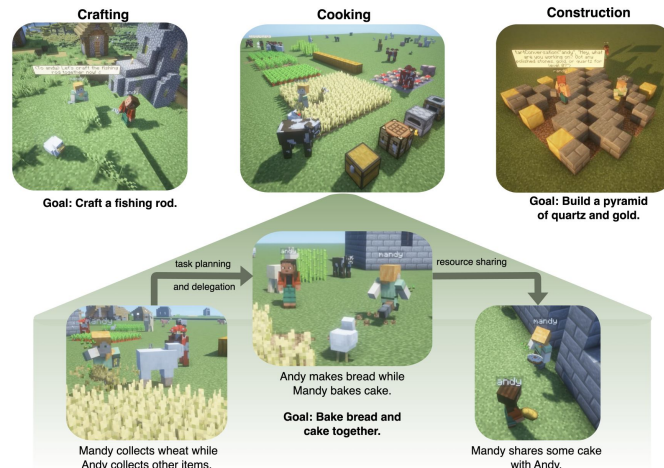


1,798 Attributes • 5,228 Objects • 23,486 Spatial Relations

Real-world Data

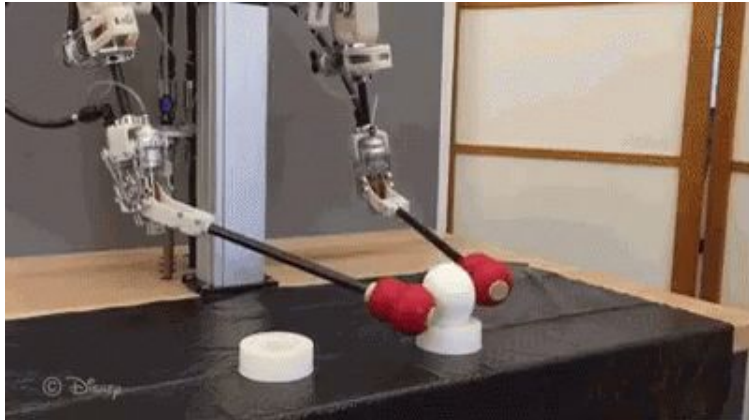
DROID
Distributed Robot
Interaction Dataset

 **76k Episodes**
 **564 Scenes**
 **52 Buildings**
 **13 Institutions**
 **86 Tasks / Verbs**

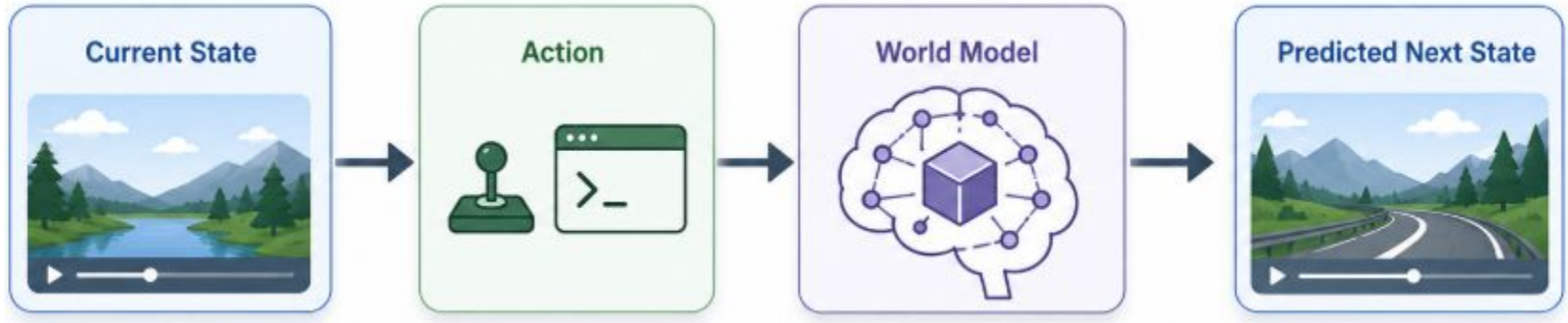


Simulated Data

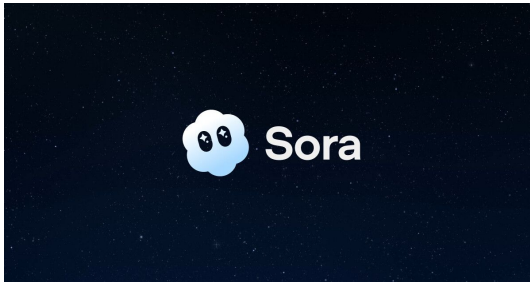
Safety Problems



World Model - A model that understand the world



Applications

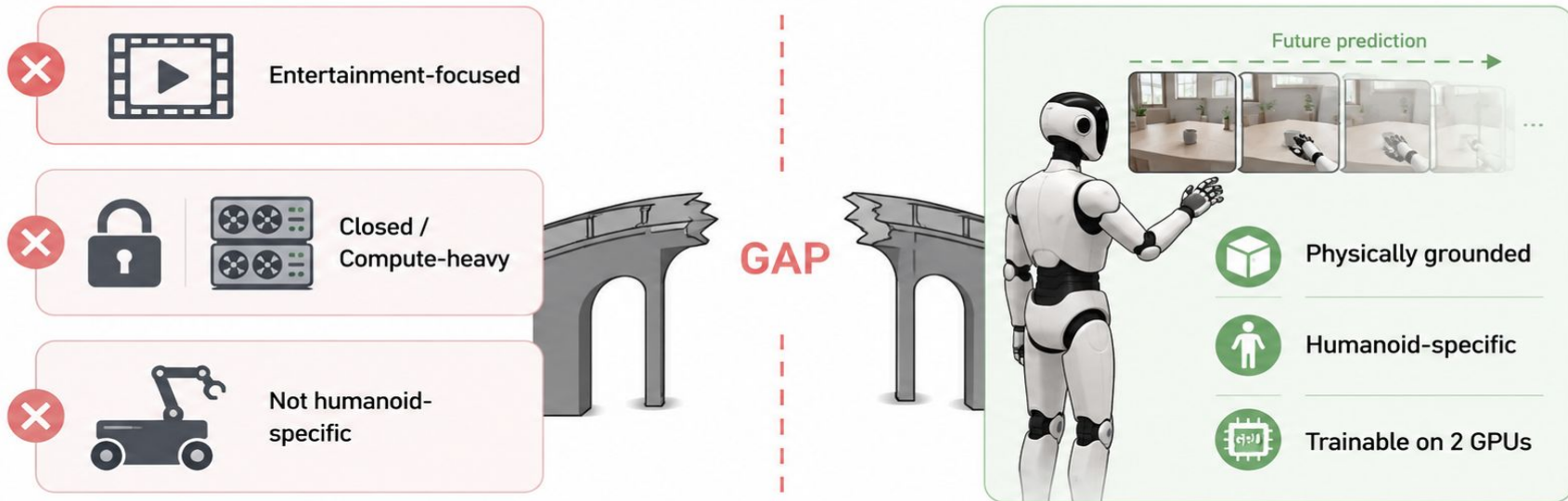


Video Generation



Games (Genie 3)

Research GAP

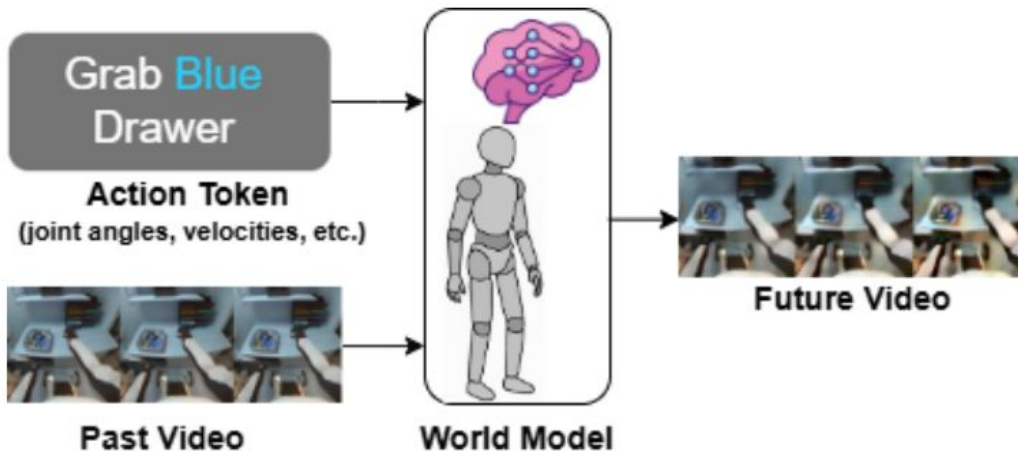


Need lightweight open world models for humanoid robotics.

Introduction to HWM



Humanoid World Models (HWM) that forecast future egocentric video conditioned on humanoid control tokens

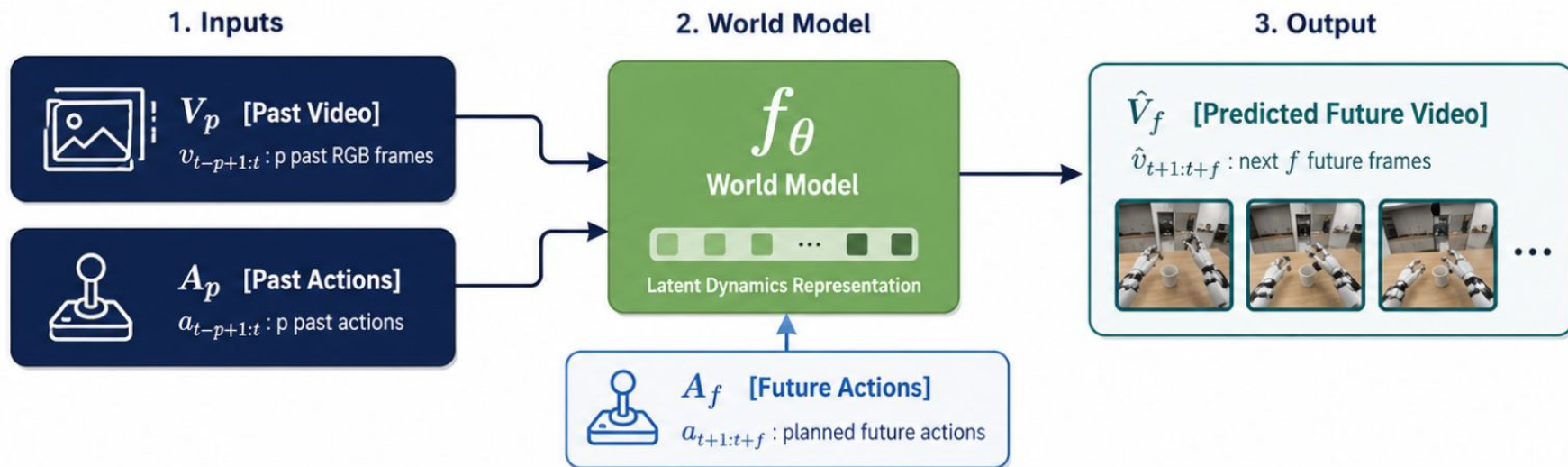


Problem Formulation

At each time step t , condition on past observations and planned actions to predict future egocentric frames.

$$\text{Learn } f_{\theta} : (V_p, A_p, A_f) \rightarrow \hat{V}_f$$

$$V_p = v_{t-p+1:t} \quad A_p = a_{t-p+1:t} \quad A_f = a_{t+1:t+f}$$



**How do we generate
future video?**

Masked Video Modelling

Future Prediction as Token Filling

Inputs

Past video V_p
Past actions A_p
Future actions A_f

VQ-VAE Tokenizer

Compress video into
discrete latent tokens
 $V_p, V_f \rightarrow L_p, L_f$

Mask Future Latents

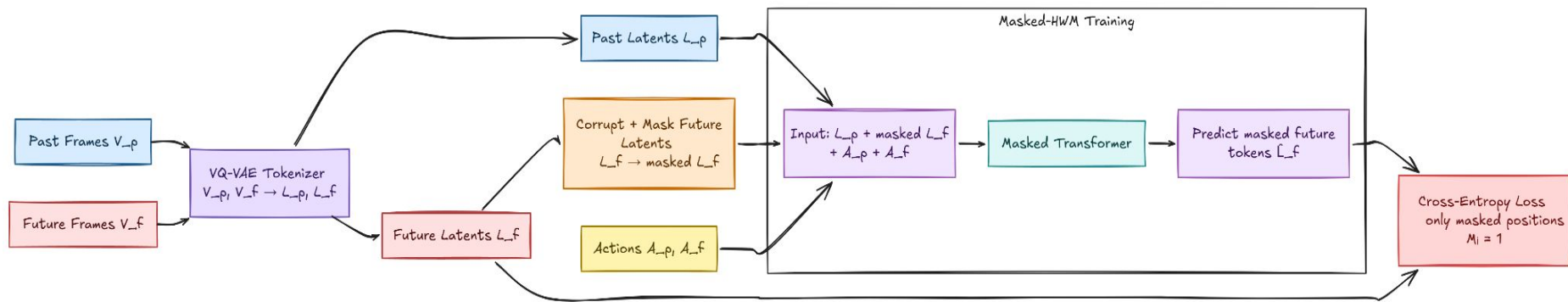
Mask future
tokens in L_f

Masked Transformer

Predict masked future
tokens using L_p, A_p and A_f

Decode

Recover future video
 $\hat{L}_f \rightarrow \hat{V}_f$



Training Objective and Inference

Original L_f

12

48

7

91

5

23

Masked input

12

[M]

7

[M]

5

[M]

Model output

12

48

7

91

5

23

Only the masked positions contribute to the loss.

Cross-Entropy Loss:

$$\mathcal{L} = -\mathbb{E}_{\mathbf{M}} \left[\sum_i \mathbf{M}_i \log p(\hat{L}_i | \mathbf{L}_f) \right]$$

$\mathbf{M}_i = 1$ only for masked positions in future latents L_f .

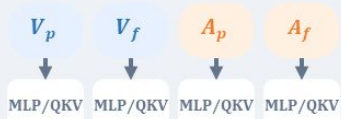
Inference:

Mask all future tokens L_f , predict tokens in parallel, then refine low-confidence tokens over K steps.

Efficiency via Parameter Sharing

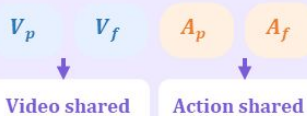
Base Block

Separate parameters for each stream



Modality Sharing

Share within similar modalities



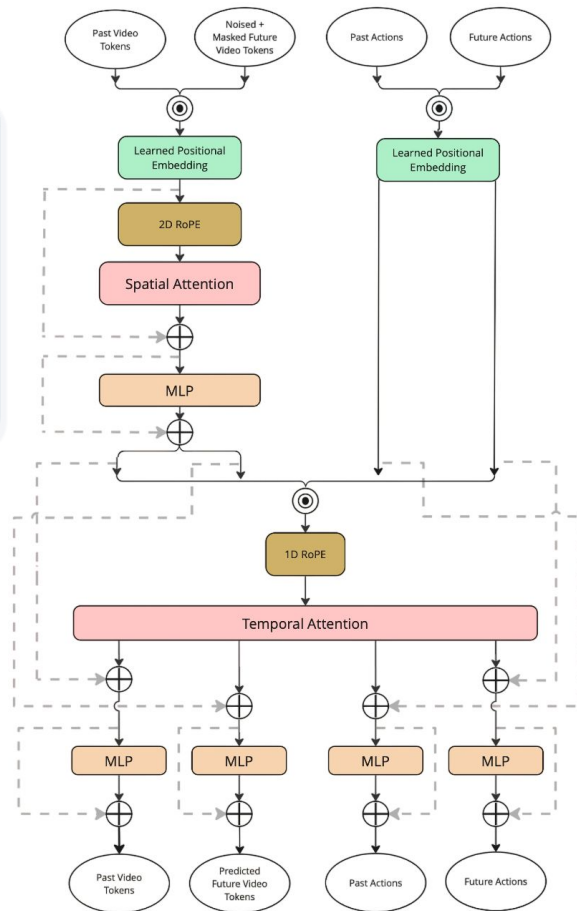
Full Sharing

Share major components across all streams



Masked-HWM parameter count

Base		0.321B	
Modality		0.237B	-26%
Full		0.195B	-39%

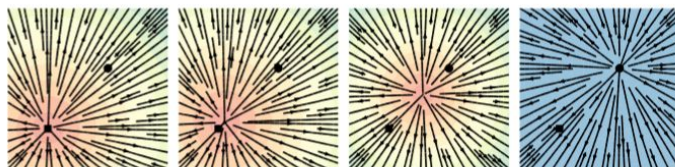
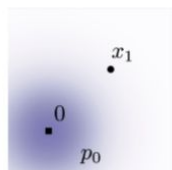
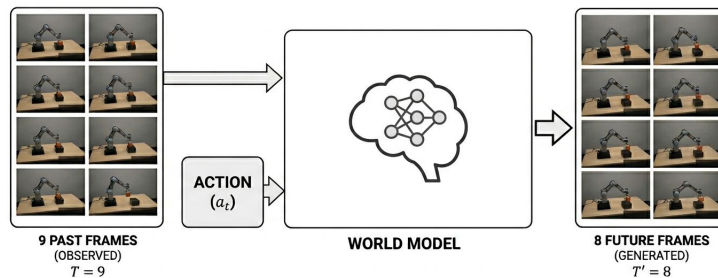
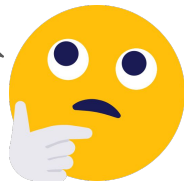


Base Block

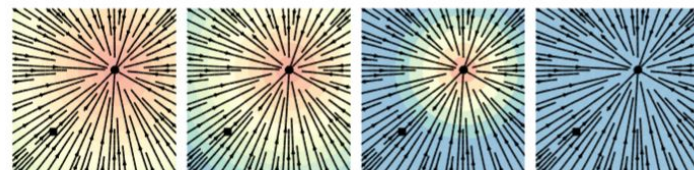
Flow Matching

Why do we need Flow Matching?

Diffusion too slow!!
Alternative??



Diffusion path – conditional score function



OT path – conditional vector field

Core Idea: Velocity Field

Line from noise to real data

$$X_t = t \cdot X_1 + (1 - (1 - \sigma_{min})t) \cdot X_0, \quad t \in [0, 1]$$

Velocity

$$V_t = \frac{dX_t}{dt} = X_1 - (1 - \sigma_{min}) \cdot X_0 \approx X_1 - X_0$$

Model learn to predict

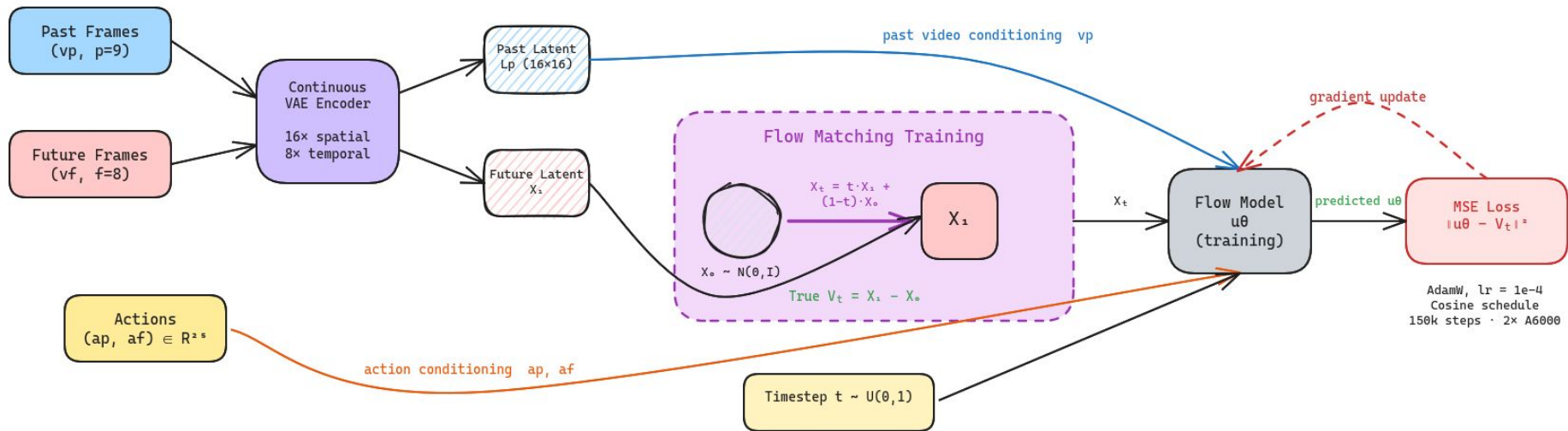
$$u_\theta(X_t, v_p, a_p, a_f, t) \approx V_t$$

With the training loss

$$\mathcal{L} = \mathbb{E} \left[\|u_\theta(X_t, v_p, a_p, a_f, t) - V_t\|^2 \right]$$

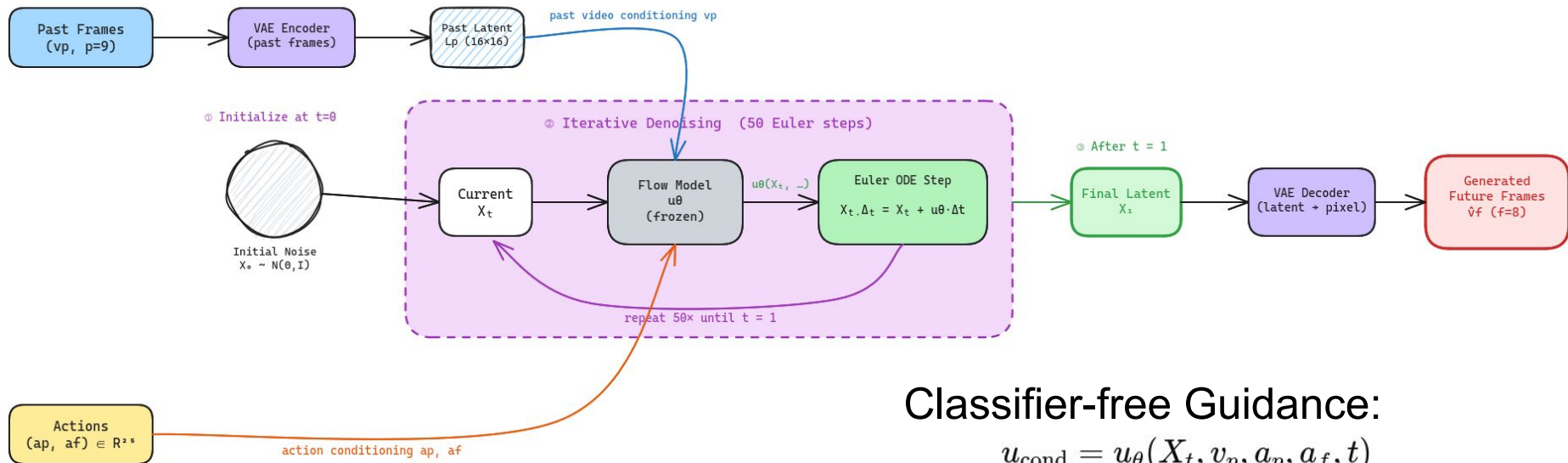
Training & Inference

Flow-HWM Training Pipeline



Training & Inference

Flow-HWM Inference Pipeline



Classifier-free Guidance:

$$u_{\text{cond}} = u_\theta(X_t, v_p, a_p, a_f, t)$$

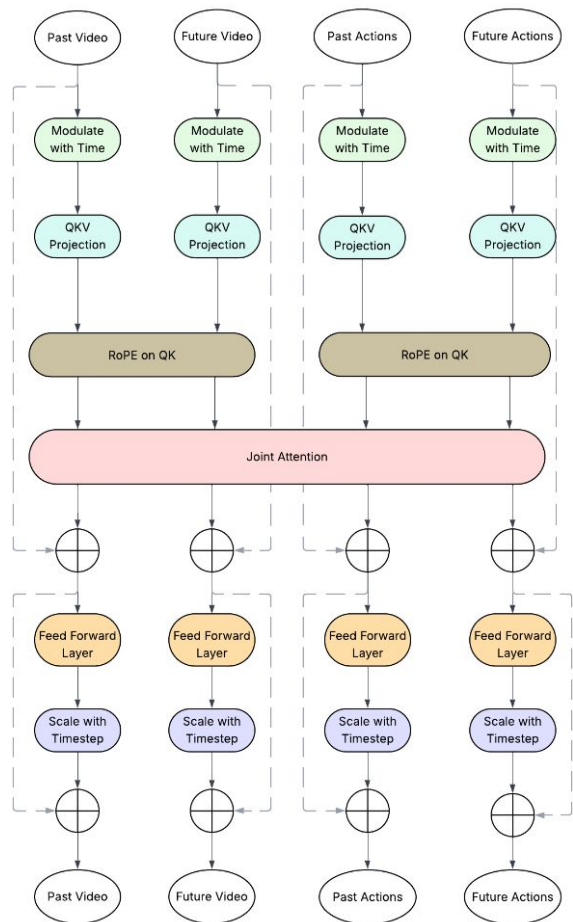
$$u_{\text{uncond}} = u_\theta(X_t, \emptyset, \emptyset, \emptyset, t)$$

$$u_{\text{guided}} = u_{\text{uncond}} + 3.0 \cdot (u_{\text{cond}} - u_{\text{uncond}})$$

$$X_{t+\Delta t} = X_t + u_{\text{guided}} \cdot \Delta t$$

Base Block of Flow-HWM

- Each token stream (past/future video and actions) uses separate weights for timestep modulation, QKV projection, and feedforward MLPs.
- Joint Attention integrates all streams.
- RoPE is applied by modality: 3D for video tokens, 1D for action tokens.



Impacts

Four core impacts:

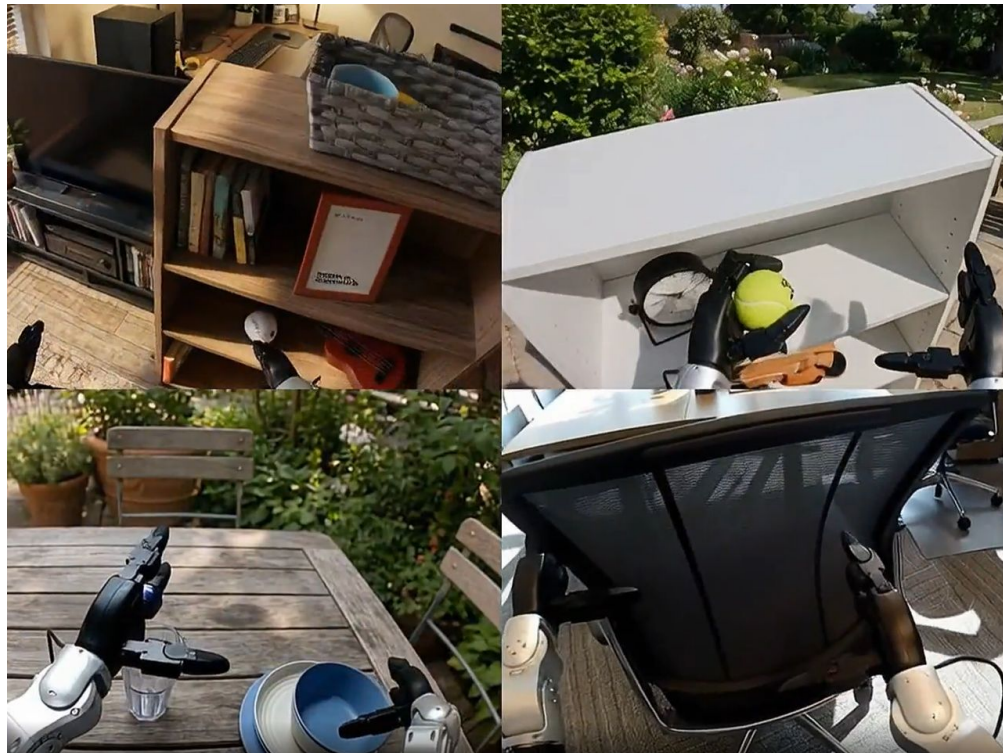
1. Democratizing AI research via resource optimization
2. Bridging the Sim-to-Real gap with synthetic data
3. Pioneering action-conditioned video generation in robotics
4. World Model as An Environment for Policy Evaluation

Democratizing Research & Optimizing Resources

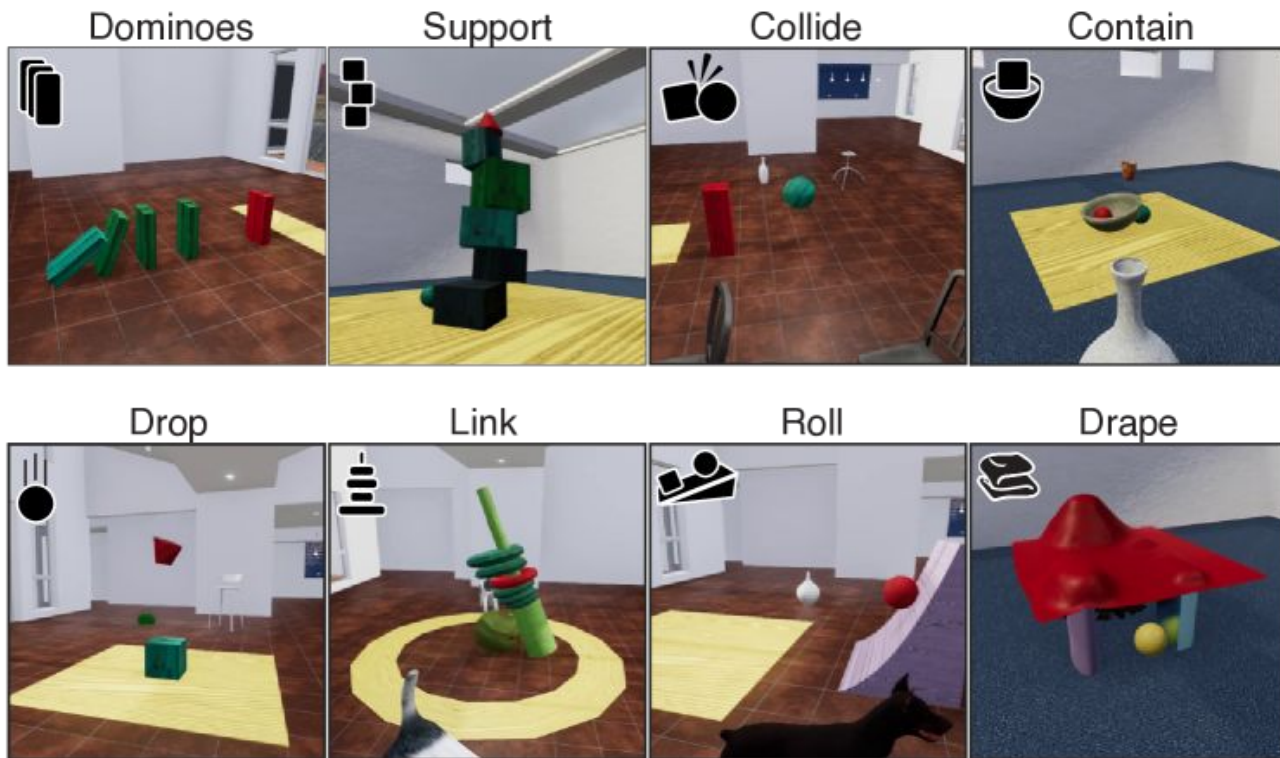
METRIC	SPLIT ATTENTION	BASE BLOCK	MODALITY SHARING	FULL SHARING
MODEL SIZE (BILLION)	0.220	0.321	0.237	0.195
PEAK GPU MEMORY (GB)	2.22	2.63	2.30	2.12
SAMPLES PER SECOND	2.09	2.27	2.25	2.36
FID	15.31	10.13	11.67	14.21
PSNR (DB)	29.37	29.02	28.97	28.66

METRIC	SPLIT ATTENTION	BASE BLOCK	MODALITY SHARING	FULL SHARING
MODEL SIZE (BILLION)	0.944	1.36	0.886	0.648
PEAK GPU MEMORY (GB)	4.37	5.94	4.41	3.25
SAMPLES PER SECOND	1.11	1.69	1.89	1.91
FID	111.12	111.59	112.75	110.73
PSNR (DB)	20.50	20.42	20.50	20.43

Bridging the Sim-to-Real Gap



Shaping the Future of Video Generation



World Model as An Environment for Policy Evaluation



Gripper Control



Sweep X-Axis



Sweep Y-Axis



Sweep Z-Axis

<https://world-model-eval.github.io/>

Our Scope

- Reimplement and optimize **Masked-HWM** for resource-constrained environments.
- Train the model on the original **1XGPT dataset**.
- Develop checkpoint loading, model export, and inference pipelines.
- Preprocess the **DROID** and **Bridge** datasets for fine-tuning and evaluation.
- Benchmark pretrained and fine-tuned models through quantitative and qualitative analyses.

Why choose Masked-HWM?

The paper reports stronger practical performance for Masked-HWM than Flow-HWM

Masked-HWM example

Generated



Ground Truth



Flow-HWM example

Generated



Ground Truth



Reported quantitative comparison

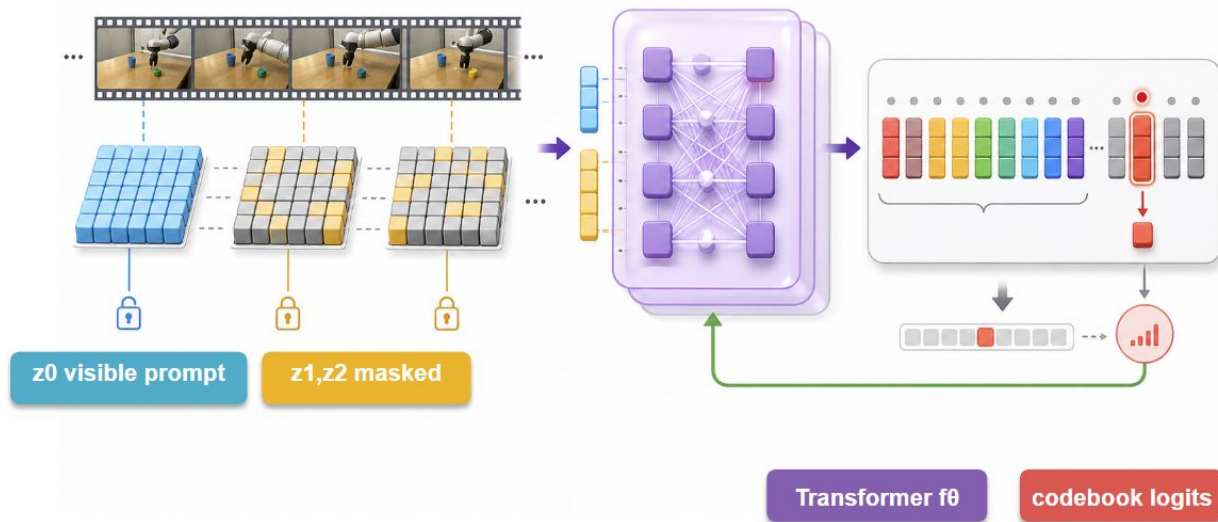
Variant	FID ↓	PSNR ↑	Params	Samples/s
Masked-HWM Base	10.13	29.02	0.321B	2.27
Masked-HWM Sharing	14.21	28.66	0.195B	2.36
Flow-HWM Base	111.59	20.42	1.36B	1.69

Masked-HWM delivers substantially better visual quality with fewer parameters and faster generation than Flow-HWM. Its discrete-token formulation is also more feasible to train and fine-tune under our limited Kaggle compute budget.

Implementation Details

Aspect	Paper / original setting	Our reimplementation
Tokenizer	Cosmos DV8x8x8 tokenizer	Cosmos DV8x8x8 tokenizer
Training target	Discrete VQ codebook IDs	Same: VQ codebook IDs
Temporal config	Original reports longer context/future setting	<code>T=3, num_prompt_frames=1</code>
Effective rollout	Predict future latent tokens	<code>z0</code> prompt → generate <code>z1, z2</code> → decode 17 RGB frames
Action conditioning	Video + action conditioning	<code>WITH_ACT=False</code> , visual-only
Loss	Cross-entropy on masked tokens	Same, CE on masked future tokens
Training steps	Reported 60k steps	10k-step Kaggle run / bounded runs
Hardware	A6000-scale training	Kaggle GPU session
Dataset	1X humanoid data	1X humanoid data + DROID & BridgeData V2

Training pipeline



Loss on masked future tokens

$$\mathcal{L}(\theta) = -\frac{1}{|\mathcal{M}|} \sum_{i \in \mathcal{M}} \log p_{\theta}(y_i | x_{\text{masked}})$$

$\mathcal{M}$ = masked token positions in z_1, z_2

Gradient signal

$$\frac{\partial \mathcal{L}}{\partial t_i} = p_{\theta}(\cdot | x_{\text{masked}}) - \text{onehot}(y_i)$$

Target y_i is a VQ codebook ID, not a pixel.
CE pushes probability toward the correct token ID.
Visual-only batch: WITH_ACT=False.

Dataset Setup

Dataset	Role	Type	Used for	Novel?
1X-style	in-distribution	humanoid/ego centric	train + val	No
DROID	OOD-1	real robot manipulation	fine-tune + eval	Yes
BridgeData V2	OOD-2	real robot / simulated robot	fine-tune/eval	Yes

Evaluation Protocol - Khang

Nội dung:

- Evaluate every checkpoint on every dataset.
- Token-space metrics:
 - CE loss ↓
 - top-1 token accuracy ↑
 - perplexity ↓
- Pixel-space metrics:
 - PSNR ↑
 - LPIPS ↓
 - FID ↓
- Efficiency:
 - generation time per frame
 - GPU memory nếu có log

Quantitative Results - Khang

Không cần nhiều số. (3 checkpoint × 3 dataset)

Điểm cần phân tích:

- Base model có generalize sang DROID/Bridge không?
- Fine-tune DROID có cải thiện DROID không?
- Có giảm performance trên original val không?
- ...

Checkpoint	Dataset	Loss ↓	Acc ↑	PPL ↓	PSNR ↑	LPIPS ↓	FID ↓	Time/frame ↓
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Qualitative Rollouts - Khang

Dùng ảnh từ notebook:

- Row 1: context frames.
- Row 2: ground-truth future.
- Row 3: predicted future.

Nên có 3 cột:

- Original val.
- DROID.
- Bridge.

Caption nên chỉ ra lỗi:

- Blur ở object nhỏ.
- Robot hand/finger/articulation sai.
- Temporal drift.
- Background giữ được nhưng interaction dynamics yếu.

Paper gốc cũng quan sát Masked-HWM học được scene structure và robot appendages, nhưng chi tiết nhỏ như fingers thường blur/entangled.

Ablation - Khang

Tham khảo chạy thêm MaskGIT steps = 1,2,4,8. Chỉ cần thay inference config, không train lại. Paper gốc dùng iterative refinement khi inference Masked-HWM; do đó ablation **K** có liên quan trực tiếp tới method.

Kết luận mong đợi:

- K thấp: nhanh nhưng chất lượng kém.
- K cao: chất lượng tốt hơn nhưng inference chậm hơn.
- Có trade-off quality–speed.

K	PSNR ↑	LPIPS ↓	FID ↓	Time/frame ↓
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Comparison with Paper - All

Aspect	Paper	Our reproduction
Model family	Masked-HWM + Flow-HWM	Masked-HWM only
Conditioning	video + actions	video-only nếu <code>WITH_ACT=False</code>
Dataset	100h humanoid demos	original subset + DROID + Bridge
Metrics	FID, PSNR, speed, memory	CE, acc, PPL, PSNR, LPIPS, FID, time
Main finding	Masked-HWM > Flow-HWM under constraints	test if Masked-HWM transfers to novel robot datasets

Final Takeaways & Limitations - All

Nội dung nên thảo:

Takeaways

- Reproduced the core Masked-HWM training/evaluation pipeline.
- Built multi-checkpoint × multi-dataset benchmark.
- Tested domain transfer from original HWM data to DROID and another novel dataset.
- Added extra ablation despite rubric not requiring it.

Limitations

- No full Flow-HWM reproduction.
- No architectural ablation: split attention / modality sharing / full sharing.
- Current implementation may not use action tokens.
- Novel datasets are robot manipulation datasets, not true humanoid egocentric datasets.

Strong final sentence:

“Therefore, our reproduction focuses on whether the lightweight discrete-token Masked-HWM paradigm remains usable under small-lab compute and cross-dataset domain shift.”